

HD189733b

R-0040

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_250912 · created 2026-07-09T00:47:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460930.7218 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.055
Transit depth (Rp/Rs)^2	3.52 +/- 2.07 [%]
Orbital Inclination (inc)	85.81 +/- 2.55
Airmass coefficient 1 (a1)	1.35 +/- 0.23
Airmass coefficient 2 (a2)	-0.25 +/- 0.2
Scatter in the residuals of the lightcurve fit is	16.97 %
Best Comparison Star	#2 - [230, 174]
Optimal Aperture	5.4
Optimal Annulus	9.44
Transit Duration (day)	0.068 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.055 vs 0.15534 (deviation 0.59σ)
Duration fit vs published	0.068 d vs 0.0754 d (deviation 0.27σ)
Mid-time O–C	-4.8 min
Depth SNR	3.4
Residual scatter	16.97 %

Light curve

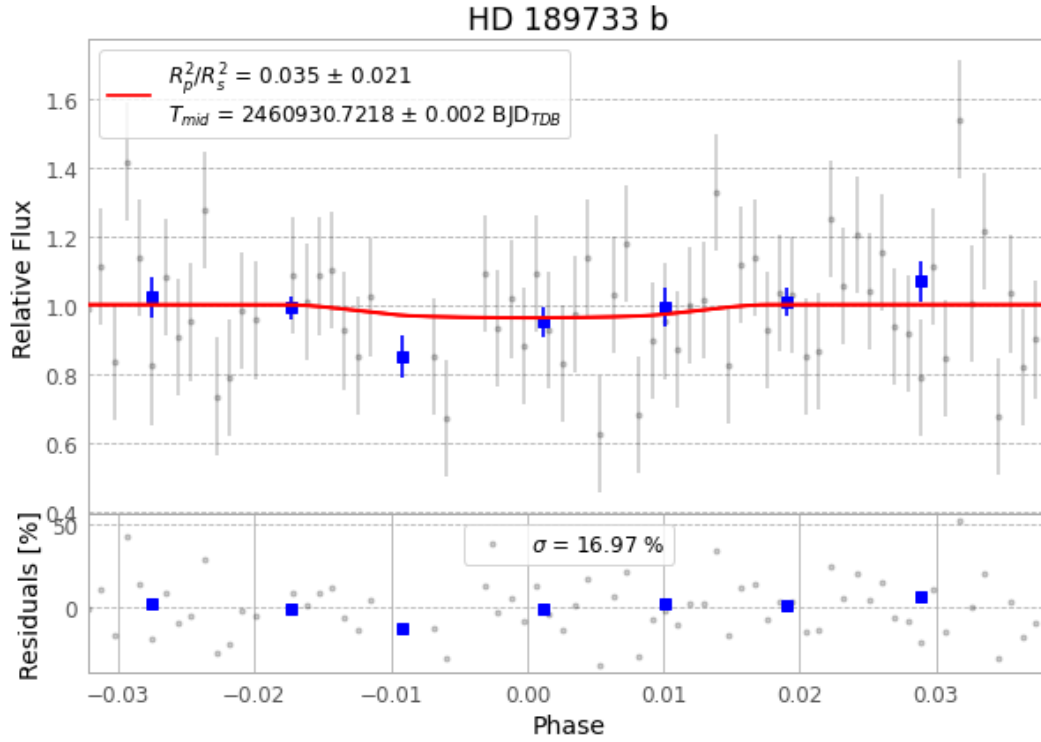


Figure 1 — FinalLightCurve_HD 189733 b_2025-09-11.png (SHA-256 a0432ecec3e91904...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 135], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6376f4e0727a5e1c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

76 FITS frames (50 MB), HD189733250912033015.FITS → HD189733250912071515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-250912.log (f07e6d1a969d...), FinalLightCurve_HD 189733 b_2025-09-11.png (a0432ecec3e9...), FinalLightCurve_HD 189733 b_2025-09-11.csv (b23614fb0163...)

Compute resources

Wall time 170.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.